

Leepaakshi Gokulkrishnan

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Education

Master of Science in Computer Science

Arizona State University

August 2023 - May 2025

GPA: 4.0/4.0

Courses: Database management, Big Data, Distributed Database Systems, Natural Language Processing, Statistical Machine Learning, Data mining, Data Structures and Algorithms

Bachelor of Technology in Computer Science

Amrita Vishwa Vidyapeetham University

August 2019 - June 2023

GPA: 3.8/4.0

Work Experience

Graduate Service Assistant

Arizona State University

August 2024 - December 2024

- Managed the final project on **hadoop, spark, and AWS**, ensuring successful completion and implementation by all students.
- Collaborated with faculty to develop instructional materials for CSE 511 **Data Processing at Scale**.

Data Engineer

Bosch Global Software

January 2023 - June 2023

- Built and optimized **ETL** workflows for navigation data of 10 TB+ with **AWS** (Glue, S3, Lambda, Redshift spectrum, Athena), cutting data prep time by 25%.
- Optimized Python/**PySpark** codebase, resulting in an impressive 96% reduction in running time.
- Developed and deployed an automated data pipeline on **AWS** (Glue, S3, and Lambda) to centralize and analyze test execution metrics across 50+ **CI/CD** pipelines.
- Built **LLM**-based test case generation pipeline using **LangChain**, **LangGraph**, and BERT embeddings, optimizing test case prioritization with Random Forest, increasing test coverage by 30%.
- Deployed **AWS EMR** clusters running **Spark** Jobs to process and analyze large-scale test log data, integrating with Cloud Watch for real-time monitoring and automated alerts, reducing test failure resolution time by 50%.

Data Engineer Intern

InterviewBit

June 2022 - January 2023

- Developed and maintained data pipelines for ingesting, transforming, and storing large datasets using **SQL**, **SSIS**, and **Python**, improving data accessibility for analytics teams.
- Built real-time interactive dashboards in **Power BI** by integrating data from **Azure Data Lake** and **Azure Synapse Analytics**, providing insights into student performance for 10,000+ students.
- Developed a deep neural network (DNN) to classify students into personalized learning paths, increasing course completion rates by 15%.
- Designed interactive **PowerBI** dashboards with DAX and KPI tracking, enhancing academic decisions through data-driven insights.

Skills

- Programming Languages & Development Frameworks:** Python, SQL, NoSQL, MySQL, PostgreSQL, Redis, MongoDB, Cassandra, Snowflake, GraphQL, Neo4j, Scala, MATLAB, Linux, Agile methodologies.
- Data Engineering & Cloud Platforms:** AWS (Glue, Lambda, S3, Redshift, DMS, Athena, Kinesis); Azure (Data Factory, SQL Database, Data Lake, Synapse); GCP
- Big Data, ML & DL:** Hadoop, PySpark, Spark, Hive, Kafka, MapReduce, RabbitMQ, LLM, Langchain, BigQuery, Bigtable, TensorFlow, PyTorch, SageMaker, HuggingFace, Caffe, Spacy, CuDA, Keras
- CI/CD & Visualization:** Terraform, Jenkins, Grafana, Docker, Kubernetes, Selenium, GitHub, Jira, PowerBI, Tableau

Projects

- Document Summarization Platform with RAG and RLHF on AWS** December 2024 - January 2024
Built a Research Document Summarization Platform using **RAG** and **RLHF** on **AWS** with **Glue**, **S3**, **OpenSearch**, **SageMaker**, and **Lambda**. The system retrieves relevant research papers, summarizes them with LLMs, and refines accuracy using human feedback, streamlining literature reviews.
- Real-time Sentiment Analysis of Tweets using Apache Spark** March 2024 - June 2024
Developed and fine-tuned a **SparkNLP** model for real-time sentiment analysis of tweets, employing Apache Spark for distributed processing and Amazon S3 for scalable storage. Leveraged **AWS Kinesis** for real-time tweet ingestion.
- Predicting Fluctuations in Cryptocurrency Value** January 2024 - June 2024
Designed an LSTM model to predict Bitcoin price fluctuations using historical price and trading volume data scraped from financial news websites. Deployed an **ETL pipeline** on **AWS EC2**, automating data ingestion and transformation processes with **Apache Airflow**, and visualized outcomes in a **PowerBI** dashboard.
- Bedtime Stories Generator** January 2024 - June 2024
Built a Web application, **React.js** frontend, **PostgreSQL** database to create personalized bedtime stories content generated by **Llama 3** model optimized using 8-bit quantization includes custom illustrations, text-to-speech function.

Certifications

AWS AI Practitioner Certification - AWS

AWS Data Engineer Associate Certification - AWS